

Project 2 – Expense Tracker

Overview

This project is part of the Decode Labs Python Programming Internship. It is a command-line application designed to help users track, organize, and manage their daily expenses efficiently.

Objective

To build a functional expense management tool that demonstrates core Python concepts such as dynamic accumulators, structured file handling with JSON, category breakdowns, and robust user input validation.

Features

Add Expenses: Log new expenses with a category and description.

Quick-Add Session: Continuous entry mode for rapidly logging multiple expenses in one go.

Running Total: Real-time summary using the accumulator pattern.

Category Breakdown: Categorized view with visual bars to track spending distribution.

Defensive Coding: Safely handles invalid inputs without crashing.

Persistent Data: Saves all financial entries to expenses.json so data persists between runs.

Concepts Used

- Accumulator variables for real-time totals
- while loop controls and sentinel values for continuous logging
- Input validation and exception handling (try/except ValueError)
- Type casting with float for monetary precision
- Data persistence using JSON reading/writing
- Modular structure and categorical data breakdown

How to Run

1. Ensure Python is installed on your system.
2. Navigate to the project directory in your terminal.
3. Run the script using the command: `python expense_tracker.py`
4. Follow the interactive menu prompts to log and view your expenses.

Future Enhancements

- Graphical User Interface (GUI) implementation using tkinter or PyQt.
- Date ranges, monthly budgeting limits, and alert thresholds.
- Exporting financial summaries to CSV or PDF reports.
- Relational database integration (SQLite) for high-volume transactions.

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